

HOMEWORK 2
BIOSTATISTICS 825
DUE SEPTEMBER 30TH, 2025

Instructions

- You may use R and any packages introduced in class (e.g., `glmnet`, `ncvreg`, `spls`, `mgcv`, `MASS`).
 - For each applied question, include both code and interpretation of results.
 - Be concise but clear: a few well-written sentences with tables/plots is preferred over long text dumps.
 - When asked to “compare models,” focus on prediction error, interpretability, and variable selection.
 - Submit your assignment as a Word file, or a knitted PDF or HTML file from R Mark-down. Include your name in the filename of your solutions.
1. **(25 points)** For this exercise we’ll use the “Communities and Crime Data Set” ([link to info](#)). This data set contains variables related to violent crime. The goal is to predict the variable “ViolentCrimesPerPop”. The first five variables in the dataset should not be used in the regression models.
 - (a) (5 pts) Fit a standard linear regression model predicting the per-capita violent crime rate from the community-level predictors. Report adjusted R^2 and comment on the fit.
 - (b) (10 pts) Fit ridge and lasso regression models using cross-validation. Report the chosen penalty parameter λ , test error, and the number of predictors retained. Compare to the ordinary regression.
 - (c) (10 pts) Fit MCP and SCAD regression models. Compare variable selection patterns to the lasso. Do MCP/SCAD retain fewer predictors while maintaining predictive accuracy? Briefly explain.
 2. **(20 points)** The dataset “parkinsons_updrs.txt” is composed of a range of biomedical voice measurements from 42 people with early-stage Parkinson’s disease recruited to a six-month trial of a telemonitoring device for remote symptom progression monitoring. The recordings were automatically captured in the patient’s homes. Columns in the table contain subject number, subject age, subject gender, time interval from baseline recruitment date, motor UPDRS, total UPDRS, and 16 biomedical voice measures. The attributes of the data are
 - subject number - Integer that uniquely identifies each subject
 - age - Subject age

- sex - Subject gender '0' - male, '1' - female
- test_time - Time since recruitment into the trial. The integer part is the number of days since recruitment.
- motor_UPDRS - Clinician's motor UPDRS score, linearly interpolated
- total_UPDRS - Clinician's total UPDRS score, linearly interpolated

16 biomedical voice measures

- Jitter(%), Jitter(Abs), Jitter:RAP, Jitter:PPQ5, Jitter:DDP - Several measures of variation in fundamental frequency
- Shimmer, Shimmer(dB), Shimmer:APQ3, Shimmer:APQ5, Shimmer:APQ11, Shimmer:DDA - Several measures of variation in amplitude
- NHR, HNR - Two measures of ratio of noise to tonal components in the voice
- RPDE - A nonlinear dynamical complexity measure
- DFA - Signal fractal scaling exponent
- PPE - A nonlinear measure of fundamental frequency variation

Each row corresponds to one of 5,875 voice recording from these individuals. The main aim of the data is to predict the motor and total UPDRS scores ('motor_UPDRS' and 'total_UPDRS') from the 16 voice measures.

- (5 pts) Fit a partial least squares (PLS) regression predicting **motor UPDRS** from the 16 biomedical voice measures. Use cross-validation to select the number of components.
 - (5 pts) Report the cross-validated prediction error and interpret how many components seem optimal. How do the results compare to lasso regression for the same outcome?
 - (5 pts) Fit a group lasso regression predicting **total UPDRS** from the grouped voice predictors.
 - (5 pts) Which groups are retained? How does grouped lasso differ from ordinary lasso in interpretation?
3. **(15 points)** This question will use the *Communities and Crime* dataset from question one.
- (5 pts) Fit a GAM predicting violent crime rate using smooth functions of at least three predictors (choose variables you believe might have nonlinear effects).
 - (5 pts) Plot the smooth terms. Are any relationships clearly nonlinear?
 - (5 pts) Compare predictive performance of the GAM to a linear model with the same predictors. Which fits better?

4. (40 points) The dataset “pima.indians.diabetes3.csv” contains data from 532 females who are at least 21 years of age and of Pima Indian heritage. The dataset consists of:

npregnant: Number of times pregnant
glucose: Plasma glucose concentration a 2 hours in an oral
glucose tolerance test
diastolic.bp: Diastolic blood pressure (mm Hg)
skinfold.thickness: Triceps skin fold thickness (mm)
bmi: Body mass index (weight in kg/(height in m)²)
pedigree: Diabetes pedigree function
age: Age (years)
classdigit: Numerical class variable
class: Alphanumeric class variable

The purpose of this study was to determine whether the patient exhibited signs of diabetes mellitus according to World Health Organization criteria (i.e., if the 2-hour post-load plasma glucose level was at least 200 mg/dL at any survey examination or if it was detected during routine care). The two class variables indicate diabetes status using a blood test of 2-hour serum insulin. The goal is to use the other variables to predict diabetes status, as obtaining information on them is easier for patients, less invasive, and less expensive than a blood test. Use all other variables in each analysis.

Split the data into training (80%) and test (20%) sets

- (a) (5 pts) Fit a logistic regression with all predictors. Perform stepwise selection by AIC and BIC. Which predictors remain in each model?
- (b) (5 pts) Fit a logistic lasso model with cross-validation to select λ . Report the number of predictors retained with the “min” and “1se” λ values.
- (c) (10 pts) Using the test data, compare the predictive ability of the models found with AIC, BIC, and lasso in terms of AUC, specificity, sensitivity, PPV, and NPV. Which approach seems best in terms of prediction and interpretability?
- (d) (5 pts) Fit an LDA model and a QDA model to classify diabetes status.
- (e) (5 pts) Use cross-validation to estimate AUC, specificity, sensitivity, PPV, and NPV for each method.
- (f) (5 pts) Compare the performance of LDA, QDA, and logistic regression. Which would you prefer in practice, and why?
- (g) (5 pts) In the description of this study, it sounds like they are more interested in using this algorithm as a *screening* tool. Does this change which approach you think is best? Why or why not?